

It is called the *mean-variance frontier*. (The next chapter will delve deeper into the economics of mean-variance frontiers for multiple assets.) In this example, the mean-variance frontier represents all the possible risk-return combinations that we can obtain by holding stock and bond positions (without employing leverage).¹⁷

Bonds lie on an *inefficient* part of the mean-variance frontier. By holding some equities, we move in a clockwise direction and strictly increase the portfolio's expected return while lowering the portfolio's volatility. Thus, we would never hold a portfolio of 100% bonds. Equity diversifies a bond portfolio because equities have a low correlation with bonds; the correlation of equities with bonds in this sample is just 11%. Thus, the investor optimally will hold some equities because equities have a chance of paying off when bonds do poorly. But how much equity should the asset owner hold?

The optimal portfolio holdings depend on the asset owner's degree of risk aversion. Panel B of Figure 2.8 overlays indifference curves for an investor with risk aversion $\gamma = 2$. She wishes to maximize her utility:

$$\max_w E(r_p) - \frac{\gamma}{2} \text{var}(r_p), \quad (2.6)$$

where we choose the optimal portfolio combination w . For this problem, specifying a weight in one asset (equities) automatically determines the weight in the other asset (bonds) as the weights sum to one. Panel B of Figure 2.8 visually depicts the optimization problem in equation (2.6). Maximizing utility is equivalent to finding the highest possible indifference curve.

The highest indifference curve is tangent to the mean-variance frontier in Panel B at the triangle. At this point, the asset owner holds 78% in equities. There are higher indifference curves graphed, but these indifference curves do not intersect the mean-variance frontier—these indifference curves are *infeasible*. We have to lie on the mean-variance frontier as these represent the only set of portfolios that can be obtained by holding positions in equities and bonds. Thus, the triangle is the highest utility that can be obtained.

¹⁷For two assets, the mean-variance frontier is traced by taking all combinations of the two assets A and B. Denoting the mean and volatility of asset A by $E(r_A)$ and σ_A , respectively, the mean and volatility of asset B by $E(r_B)$ and σ_B , respectively, and the correlation between the returns of asset A and asset B by ρ , the portfolio mean and volatility for the mean-variance frontier for two assets are given By

$$E(r_p) = wE(r_A) + (1 - w)E(r_B)$$

$$\sigma_p = \sqrt{w^2\sigma_A^2 + (1 - w)^2\sigma_B^2 + 2w(1 - w)\rho\sigma_A\sigma_B},$$

where we vary the portfolio weight w held in asset A. The charts in Figure 2.8 restrict w to lie between 0 and 1.